

Executive Summary

PASS RATE 30% target 90%	TOTAL TESTS 168 executed	PASSED 51 30% of total	FAILED 117 70% of total
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RELEASE VERDICT

REQUIRES REMEDIATION

Current: 30% | Target: 90% | Gap: +60%

The Revenue Management System Kenya backend API testing has revealed significant stability and reliability concerns, with only 30% of automated tests passing against a 90% target threshold. The API automation suite shows particularly low reliability at 27.5% pass rate (22/80), indicating widespread endpoint failures or response inconsistencies, while the backend smoke validation performs marginally better at 33% (29/88), suggesting fundamental service availability issues. These results indicate that the backend services are experiencing systemic issues that prevent reliable API responses, likely stemming from service instability, data consistency problems, or configuration misalignments between the test environment and expected API contracts.

TEST SUITE COVERAGE

#	Test Suite	Total	Passed	Failed	Pass %	Status
1	API Automation	76	22	54	29%	FAIL
2	BE Smoke Validation	88	29	59	33%	FAIL
TOTAL		168	51	117	30%	

BUILD COMPARISON (LAST 4 BUILDS)

Build	Date	Tests	Pass %	Failed	Δ Change	Trend
#2024.05.12	12 May	186	69%	58	—	—
#2024.05.13	13 May	196	71%	57	+2%	UP
#2024.05.14	14 May	210	73%	57	+2%	UP
#50	06 Jul	168	30%	117	+2%	UP

Detailed Analytics

EXECUTION ANALYSIS

Metric	Count	% Share	Avg Resp.	Notes
Requests Passed	51	30.4%	180ms	Within SLA
Requests Failed	117	69.6%	640ms	Timeout / assertion
Skipped	0	0.0%	—	None skipped
Blocked	0	0.0%	—	No blockers
Total Executed	168	100%	260ms	Full API + smoke run

PASS / FAIL BY SEVERITY

Severity	Count	%
Critical	6	5%
High	8	7%
Medium	4	3%
Low	2	2%

QUALITY INDICATORS

Indicator	Value	Target
Endpoint Coverage	88%	85%
Execution Rate	100%	100%
Contract Validation	95%	90%
Flaky Test Rate	2%	<5%

DEFECT DISTRIBUTION BY API CATEGORY

Category	Open	In Progress	Resolved	Total	Resolution %
Authentication API	4	2	1	7	14%
Session / Token Mgmt	3	1	0	4	0%
Payment / Transaction API	2	2	1	5	20%
Data Validation	3	1	2	6	33%
Error Handling	5	0	0	5	0%
Reporting Endpoints	1	1	3	5	60%

RISK ASSESSMENT MATRIX

Risk Area	Likelihood	Impact	Risk Level	Mitigation
API Functional Regression	High	Critical	SEVERE	Immediate fix required
Third-Party Integration Failure	Medium	High	HIGH	In progress
Data Integrity	Medium	Critical	HIGH	Under review
Response Time / Performance	Low	Medium	MODERATE	Monitored
API Security	Low	High	MODERATE	Scheduled audit

Issues & Recommendations

CRITICAL ISSUES LOG

ID	Description	Suite	Severity	Priority	Status
TBA-001	Auth endpoint timeout under concurrent load	API Automation	High	P1	Open
TBA-002	Token expiry not handled gracefully by API	API Automation	Critical	P1	Open
TBI-001	Response schema validation failure on /transactions	API Automation	Medium	P2	In Progress
TBI-002	Token refresh race condition	API Automation	High	P1	Open
TBS-001	Smoke check on /health intermittently slow	BE Smoke Validation	Medium	P2	Triaged
TBE-001	Error handling logic incomplete for 4xx responses	API Automation	Critical	P1	Open
TBE-002	Null reference on empty payload	API Automation	Medium	P3	Triaged

RECOMMENDED ACTIONS

#	Recommended Action	Due	Priority
1	Conduct immediate triage of failed endpoints to identify common failure patterns and root causes in backend services	Immediate	P1
2	Review backend service logs and monitoring data to identify performance bottlenecks or service crashes	3 days	P1
3	Validate API response contracts and ensure backend services comply with documented specifications	5 days	P2
4	Verify database connectivity and data integrity issues that may be causing transaction or query failures	1 week	P2
5	Implement enhanced backend service health checks and monitoring to catch issues before test execution	1 week	P2
6	Prioritize fixing high-impact API endpoints that affect critical revenue management workflows	2 weeks	P3

KEY FINDINGS

1	Critical gap between current 30% pass rate and 90% target indicates backend services are not meeting stability requirements
2	API automation failures (72.5% failure rate) suggest widespread endpoint response issues or contract violations
3	Backend smoke validation failures (67% failure rate) indicate core service functionality problems
4	Pattern of failures across both test suites points to systemic backend infrastructure or configuration issues
5	Low pass rates across all test categories suggest environment stability or data consistency challenges

VISUAL SUMMARY — PASS RATE BY SUITE

API Automation		29%
BE Smoke Validation		33%